
HELICOPTER ASSIGNMENT

AE4314-21 Helicopter Performance, Stability and Control

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List of Symbols

Symbol	Description
g	Acceleration due to gravity [m/s ²]
ρ	Air density [kg/m ³]
$MTOW$	Maximum Take-Off Weight [kg]
W	Weight of the helicopter [N]
T	Rotor thrust [N]
N	Number of rotor blades [–]
R	Rotor radius [m]
c	Blade chord length [m]
Ω	Rotor angular velocity [rad/s]
v_i	Induced velocity at the rotor disc [m/s]
$v_{i,hov}$	Induced velocity in hover [m/s]
V	Helicopter forward speed [m/s]
μ	Advance ratio, defined as $\mu = \frac{V}{\Omega R}$ [–]
α_d	Rotor disc angle of attack [rad]
S_{eq}	Equivalent flat plate area [m ²]
C_{D_p}	Profile drag coefficient of the rotor blades [–]
σ	Rotor solidity, defined as $\sigma = \frac{Nc}{\pi R}$ [–]
P_{ideal}	Ideal hover power from ACT [W]
P_{hov}	Actual hover power [W]
P_i	Induced power in forward flight [W]
P_p	Rotor profile drag power [W]
P_d	Rotor drag power (due to H_0 force) [W]
P_{par}	Fuselage parasite power [W]
P_{climb}	Climb power component [W]
P_{misc}	Miscellaneous power (tail rotor, gearbox, etc.) [W]
H_0	Force due to asymmetry in profile drag [N]
k	Induced power correction factor [–]
FM	Figure of Merit, efficiency factor for rotor hover performance [–]
$\dot{m}$	Mass flow rate through the rotor disk [kg/s]
β	Blade flapping angle [rad]
ψ	Azimuthal blade position [rad]
q	Pitch rate of the helicopter body [rad/s]
p	Roll rate of the helicopter body [rad/s]

M_a	Aerodynamic moment on the blade [Nm]
I_{bl}	Blade moment of inertia [kg m ²]
C_L, C_D	Lift and drag coefficients of the blade airfoil [–]
θ_0	Collective pitch angle [rad]
γ	Flight path angle [rad]
Φ	Inflow angle of relative wind at blade element [rad]

Chapter 1

Part 1 - Helicopter Performance Calculation

1.1 Choice of helicopter configuration

The helicopter chosen for this assignment is the **Eurocopter EC135P1**, also known as the Airbus Helicopters H135. This model was selected due to its popularity and widespread use. In the Netherlands alone, four are operated as trauma helicopters [9], with additional units serving in police and firefighting roles [5]. Since its introduction in 1996, the H135 has become one of the most widely used helicopters worldwide, valued for its versatility, efficiency, and low operating costs [1].

The H135 is classified as a twin-engine, light-utility civil helicopter. However, due to its adaptability and customizability, it is also employed in various military applications [2].

One of the key advantages of selecting this model for analysis is the abundance of publicly available data, which facilitates accurate estimations of its parameters. Table 1.1 provides a summary of the helicopter's key specifications:

Parameter	Value	Unit	Source
Masses			
M_{empty}	1455	kg	[2][7]
MTOW	2910	kg	[2][7]
$M_{\text{fuel, max}}$	560	kg	[2][7]
$M_{\text{Payload, max}}$	1360	kg	[2]
Performance			
Max. Disc loading	35.6	kg/m^2	[2]
Max. Rate of climb	457	m/min	[2]
Service ceiling	3050	m	[2]
Hover ceiling (IGE)	3050	m	[2]
Hover ceiling (OGE)	1830	m	[2]
Cruise speed (max.)	256	km/h	[2][7]
Cruise speed (econ.)	239	km/h	[2]
Max. Range	620	km	[2]
max. Endurance	203	min	[2][7]
Rotor			
M_{blade}^1	36.7	kg	[3][8]
Blade radius (R)	5.2	m	[2][7]
Number of blades (N)	4	—	[2][7]
Equivalent blade chord	0.288	m	[7][6]
Rotor RPM	395	RPM	[7]

Table 1.1: Key specifications of the Eurocopter EC135P1 (note that the sources can be found in the fourth column)

1.2 Induced velocity calculation

The induced velocity in hovering flight can be determined using **Actuator Disc Theory**, which models the rotor as a uniformly loaded actuator disc. The theory is based on the conservation of mass, momentum, and energy.

Derivation of Induced Velocity in Hover

To derive the induced velocity in hover, we start with the following fundamental principles:

1. Continuity Equation:

$$\dot{m} = \rho \pi R^2 v_i \quad (1.1)$$

where:

- $\dot{m}$ is the mass flow rate through the rotor disc,
- ρ is the air density,
- R is the rotor radius,
- v_i is the induced velocity at the rotor disc.

¹The mass of a single blade out of 4.

2. Momentum Equation:

The total thrust T generated by the rotor is given by:

$$T = \dot{m} \cdot 2v_i \quad (1.2)$$

Substituting $\dot{m}$ from the continuity equation:

$$T = 2\rho\pi R^2 v_i^2 \quad (1.3)$$

3. Solving for Induced Velocity: In hovering flight, the total thrust T is equal to the weight W of the helicopter, so:

$$W = 2\rho\pi R^2 v_i^2 \quad (1.4)$$

Solving for v_i , we obtain the induced velocity in hover:

$$v_{ihov} = \sqrt{\frac{W}{2\rho\pi R^2}} \quad (= 11.94\text{m/s}) \quad (1.5)$$

Using this equation, the induced velocity in hovering flight, assuming maximum take-off weight ($W_{MTOW} = 28547N$), can be determined as $v_{ihov} = 11.94\text{m/s}$ at ground level. This can also be observed in Figure 1.1b, where $V_\infty = 0$.

Derivation of Induced Velocity in Forward Flight

Next, this analysis can be extended to include forward flight. Here, The velocity through the rotor disc can be expressed by the induced velocity v_i and the free stream velocity (V_∞). An equation for the rotor thrust can thus be set up in a similar manner as in hovering flight:

$$\dot{m} = \rho\pi R^2 V_R \quad (1.6)$$

$$T = \dot{m} \cdot 2v_i \quad (1.7)$$

in Equation 1.6, it can be seen that the mass flow is a function of the rotor velocity (V_R) while the thrust is still only dependent on the change in velocity (v_i) as can be seen in Equation 1.7. Combining these two equations yields the following expression for v_i , called the Glauert formula:

$$v_i = \frac{T}{2\rho\pi R^2 V_R} \quad (1.8)$$

The velocity through the rotor can now be expressed as the perpendicular- and tangential components of the freestream velocity on the disc along with the induced velocity:

$$V_R = \sqrt{(V_\infty \cos(\alpha_d))^2 + (V_\infty \sin(\alpha_d + v_i))^2} \quad (1.9)$$

Substituting Equation 1.8 into Equation 1.9 and squaring it yields the following equation for v_i :

$$v_i^2 = \left(\frac{T}{2\rho\pi R^2} \right)^2 \frac{1}{(V_\infty \cos(\alpha_d))^2 + (V_\infty \sin(\alpha_d + v_i))^2} \quad (1.10)$$

Nondimensionalizing the equation by the induced velocity in hover:

$$v_{i,h} = \sqrt{\frac{T}{2\rho\pi R^2}} \quad ; \quad \bar{V} = \frac{V_\infty}{v_{i,h}} \quad ; \quad \bar{v}_i = \frac{v_i}{v_{i,h}}$$

$$\implies \bar{v}_i^2 = \frac{1}{(\bar{V} \cos(\alpha_d)^2 + (\bar{V} \sin(\alpha_d) + \bar{v}_i)} \quad (1.11)$$

In order to solve for the induced velocity for a given airspeed, first the disc angle of attack (α_d) must be determined; this can be done by solving the trim condition parallel to the airspeed direction given by Equation 1.12:

$$T \sin(\alpha_d) - D_{fus} - W \sin \gamma = 0 \quad (1.12)$$

Fuselage drag can be estimated by Equation 1.13 where $\sum(C_D S)_S$ is the equivalent flat plate area, which can be estimated from statistical data based on the gross weight. For this assignment, the velocity calculations were done for MTOW, so this was taken as the gross weight, yielding an equivalent flat plate area of $21.5 \text{ ft}^2 = 2.00 \text{ m}^2$.

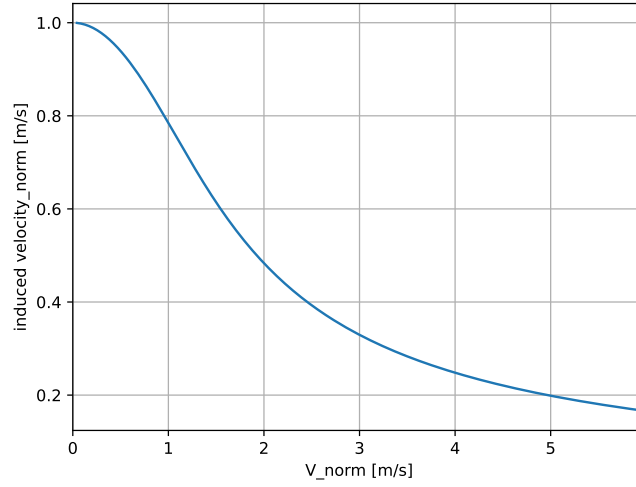
$$D_{par} = \sum(C_D S)_S \frac{1}{2} \rho V^2 \quad (1.13)$$

Assuming level flight (i.e. $\gamma = 0$, and thus $\sin(\gamma) = 0$). This also means that the thrust is equal to $W/\cos(\alpha_d)$. Using this, Equation 1.12 can thus be reduced to:

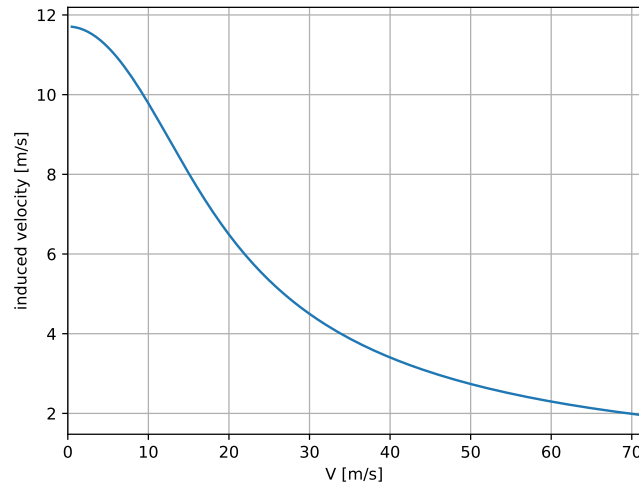
$$\frac{T}{W} \sin(\alpha_d) - \frac{D_{fus}}{W} = 0 = \frac{\sin(\alpha_d)}{\cos(\alpha_d)} - \frac{D_{fus}}{W}$$

$$\implies \alpha_d = \arctan\left(\frac{D_{fus}}{W}\right) \quad (1.14)$$

With the drag, and the weight of the helicopter known, the disc angle of attack can be determined and thus Equation 1.11 can be solved numerically for the given α_d at any velocity. This is done by calculating the hover-induced velocity $v_{i,h}$ with $T = W/\cos(\alpha_d)$, then normalizing the flight velocity and inserting it into Equation 1.11 which outputs the normalized induced velocity numerically. This is plotted in Figure 1.1a. The normalized induced velocity is then multiplied with $v_{i,h}$ to find the absolute induced velocity for all level forward flight velocities within the flight regime of the EC135P1 as found in Table 1.1. This is plotted in Figure 1.1b



(a) $\bar{v}_i$ against $\bar{V}_\infty$ for the EC135P1 at MTOW in level flight



(b) v_i against V_∞ for the EC135P1 at MTOW in level flight

Figure 1.1: Comparison of induced velocity representations for the EC135P1 at MTOW

1.3 Performance calculations

Hover Calculations

Ideal Power

The ideal power required for a helicopter in hover is derived from the Actuator Disk Theory (ACT). As derived in the lectures, the power required can be expressed as:

$$P_{id} = W \sqrt{\frac{W}{2\rho\pi R^2}} \quad (= 285.3\text{kW}) \quad (1.15)$$

where: W is the weight of the helicopter, ρ is the air density and R is the rotor radius.

This expression represents the minimum power required to sustain hover, assuming an ideal rotor without losses.

Power Calculation using Actuator Disk Theory (ACT) and Blade Element Momentum (BEM) Theory

To determine the power required for hover using ACT and BEM theory, we will incorporate the Figure of Merit (FM) to account for rotor efficiency. The FM of the H135 results from its aerodynamic characteristics and losses in the system.

The power required in ACT is given by:

$$P_{\text{hov}} = \frac{P_{\text{id}}}{FM} \quad (= 475.5\text{kW, with FM} = 0.6) \quad (1.16)$$

For the BEM theory, the power calculation involves integrating the aerodynamic forces along the rotor blade, considering induced velocity and profile drag. This can be computed using the following formula:

$$P_{\text{hov}} = P_i + P_p = kTv_i + \frac{\sigma \overline{C}_{D_p}}{8} \cdot \rho(\Omega R)^3 \pi R^2 \quad (= 328.1 + 109.5 = 437.6 \text{ kW}) \quad (1.17)$$

Where P_i is the induced power required which is the power needed to overcome the blades induced drag and the profile drag power P_p is the power needed to overcome the blades profile drag.

The rotor solidity (σ) is a key parameter that represents the fraction of the rotor disk covered by the blades, defined as:

$$\sigma = \frac{Nc}{\pi R} (= 0.0705) \quad (1.18)$$

where N is the number of blades, c is the blade chord, and R is the rotor radius.

Solidity affects power consumption: lower σ improves hover efficiency by reducing drag, while higher σ enhances maneuverability in forward flight. Since profile drag power depends on solidity,

$$P_p = \frac{\sigma \overline{C}_{D_p}}{8} \cdot \rho(\Omega R)^3 \pi R^2, \quad (1.19)$$

higher σ increases P_p , leading to higher power requirements.

Forward Flight Calculations

In forward flight, the total power required for a helicopter can be decomposed into multiple components. These include induced power, profile drag power, rotor drag power, parasite power, climb power (if applicable), and miscellaneous losses. The total required power P_{total} is expressed as:

$$P_{\text{total}} = P_i + P_p + P_d + P_{\text{par}} + P_{\text{misc}} + P_{tr} + P_{\text{climb}} \quad (1.20)$$

Each term is described below.

1. Induced Power P_i

Induced power is the power required to overcome the induced drag force, or more specifically, to move a mass of air through the rotor disk at the induced velocity v_i . This power is given by:

$$P_i = \frac{N}{2\pi} \int_0^{2\pi} d\psi \int_0^R v_i dT = kP_{id} = kTv_i \quad (1.21)$$

where, k is an empirical correction factor chosen as $k = 1.15$. The induced velocity as a function of forward flight speed has been derived in section 1.2, remember, from the same section, $T = W/\cos(\alpha_d)$ with $\alpha_d = \arctan(\frac{D_{fus}(V)}{W})$.

In forward flight, P_i decreases with increasing flight speed. As helicopter goes faster, more air goes through the stream tube and the rotor has to do less work.

2. Rotor Profile Drag Power P_p

This is the power required to overcome the friction drag of the rotor blades. It increases with rotor angular speed and depends on the blade solidity σ and drag coefficient C_{D_p} :

$$P_p = \frac{\sigma C_{D_p}}{8} \rho (\Omega R)^3 \pi R^2 (1 + \mu^2) \quad (1.22)$$

Where μ is the advance ratio, which is defined as follows:

$$\mu = \frac{V \cos(\alpha_d)}{\Omega R} \approx \frac{V}{\Omega R}$$

3. Rotor Drag Power P_d

This arises due to the asymmetry in profile drag between the advancing and retreating blades, generating the so-called H_0 force. The corresponding power is:

$$P_d = H_0 V \cos(\alpha_d) \quad (1.23)$$

with H_0 given by:

$$H_0 = \frac{1}{4} \sigma \rho C_{D_p} R^2 \pi \mu^2 \quad (1.24)$$

and the advance ratio defined as $\mu = V/(\Omega R)$.

4. Parasite Power P_{par}

This is the power required to overcome drag of all the fuselage, except the rotor blades, i.e.:

$$P_{\text{par}} = D_{\text{fus}} V = \sum (C_D S)_S \frac{1}{2} \rho V^3 \quad (1.25)$$

where $\sum (C_D S)_S$ is the equivalent flat plate area of the helicopter.

5. Miscellaneous Power P_{misc}

This accounts for power used by the tail rotor, gearbox, hydraulic pumps, and electrical systems, typically around 6–8% of the required power for rotor.

6. Tailrotor Power P_{tr}

Above the power calculations derived in the lectures, the calculation will be extended with the tailrotor power. P_{itr} being the power required to overcome the induced drag of the tailrotor, defined as:

$$P_{itr} = 1.1k_{tr}T_{tr}v_{itr} \quad , \text{ where } k_{tr} = 1.4 \quad (1.26)$$

This is the only power contribution that will be considered for the tailrotor, for example, the tailrotor profile drag power is assumed to be negligible. Some of the required power that could cause by the tailrotor is assumed to be covered by P_{misc} .

Forward Flight Results and Discussion

The results of the power breakdown in forward flight using Blade Element Momentum theory are shown in Figure 1.2. The total power required is obtained by summing the induced power, profile drag power, rotor drag power, parasite power, tail rotor power, and an additional 6% miscellaneous power loss.

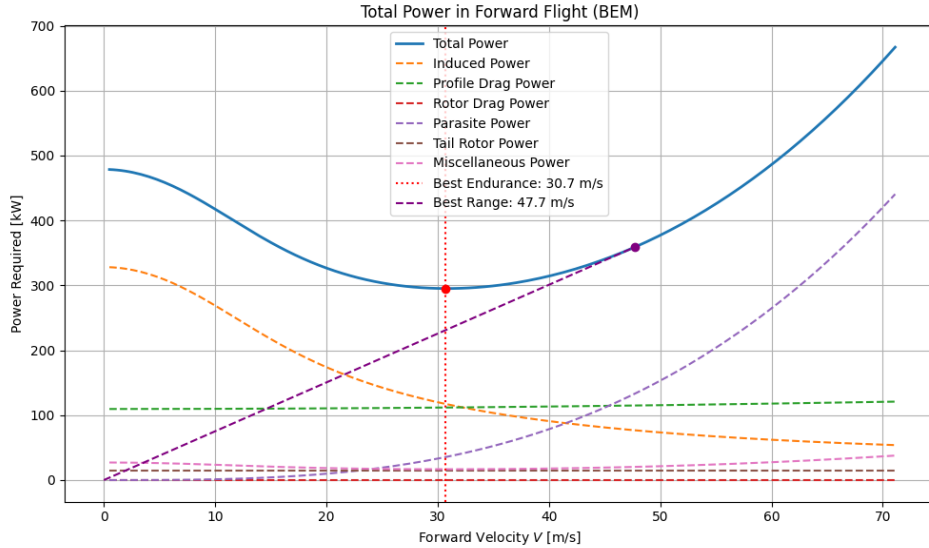


Figure 1.2: Total power required in forward flight with breakdown into components, including indications for best endurance and best range speeds.

Two key performance speeds were extracted from the power curve:

- **Best endurance speed:** This is the velocity at which the total power required is minimized. It corresponds to the point of minimum power on the curve. This speed was found to be:

$$V_{\text{endurance}} = 30.7 \text{ m/s} \quad (1.27)$$

- **Best range speed:** This is the velocity for which the power-to-speed ratio P/V is minimized, i.e., the tangent from the origin to the power curve. This corresponds to the most efficient cruising speed in terms of distance flown per unit energy. This speed was determined to be:

$$V_{\text{range}} = 47.7 \text{ m/s} \quad (1.28)$$

Both points are indicated on the plot. The best endurance point lies at the bottom of the power curve, while the best range point lies on the tangent line drawn from the origin to the curve. These velocities are lower compared to those indicated by Airbus: the best endurance speed is 239 km/h (66.4 m/s) and the best range speed is 256 km/h (71.1 m/s). This discrepancy may be caused by an unmodeled increase in power consumption at lower speeds, where, for example, the induced power could be higher than predicted. Alternatively, the parasite power might have been overestimated due to challenges in accurately estimating the helicopter's equivalent flat plate area, leading to a conservative estimate and thus a slightly excessive power requirement at higher speeds.

Chapter 2

Part 2 - Helicopter Rotor Dynamics

2.1 Determination of the coming angles

In order to solve the blade flapping motion, the complete equation of motion must be set up. In Lecture 10, the equation for forward flight with a pitch rate q is given by Equation 2.1. In order to analyse the motion described by the assignment, an additional Coriolis term must be added to account for the roll rate of the vehicle. This appended equation is given by Equation 2.2.

$$\ddot{\beta} + \Omega\beta = -2q \sin(\Psi) + \frac{M_a}{I_{bl}} \quad (2.1)$$

$$\ddot{\beta} + \Omega\beta = -2q \sin(\Psi) + 2p \cos(\Psi) + \frac{M_a}{I_{bl}} \quad (2.2)$$

in these equations, M_a is the aerodynamic moment given by Equation 2.3 where α is given by Equation 2.4:

$$M_a = \int_0^R C_{l\alpha} \alpha \frac{1}{2} \rho (\Omega r + V \cos(\alpha_c) \sin(\Psi))^2 c r dr \quad (2.3)$$

$$\alpha = \theta - \frac{V \sin(\alpha_c) + v_i + \dot{\beta}r - qr \cos(\Psi) - pr \sin(\Psi) + \beta V \cos(\alpha_c) \cos(\Psi)}{\Omega r + V \cos(\alpha_c) \cos(\Psi)} \quad (2.4)$$

To better understand the terms in Equation 2.4, a visual breakdown of the relevant velocity components and angular definitions is shown in Figure 2.1.

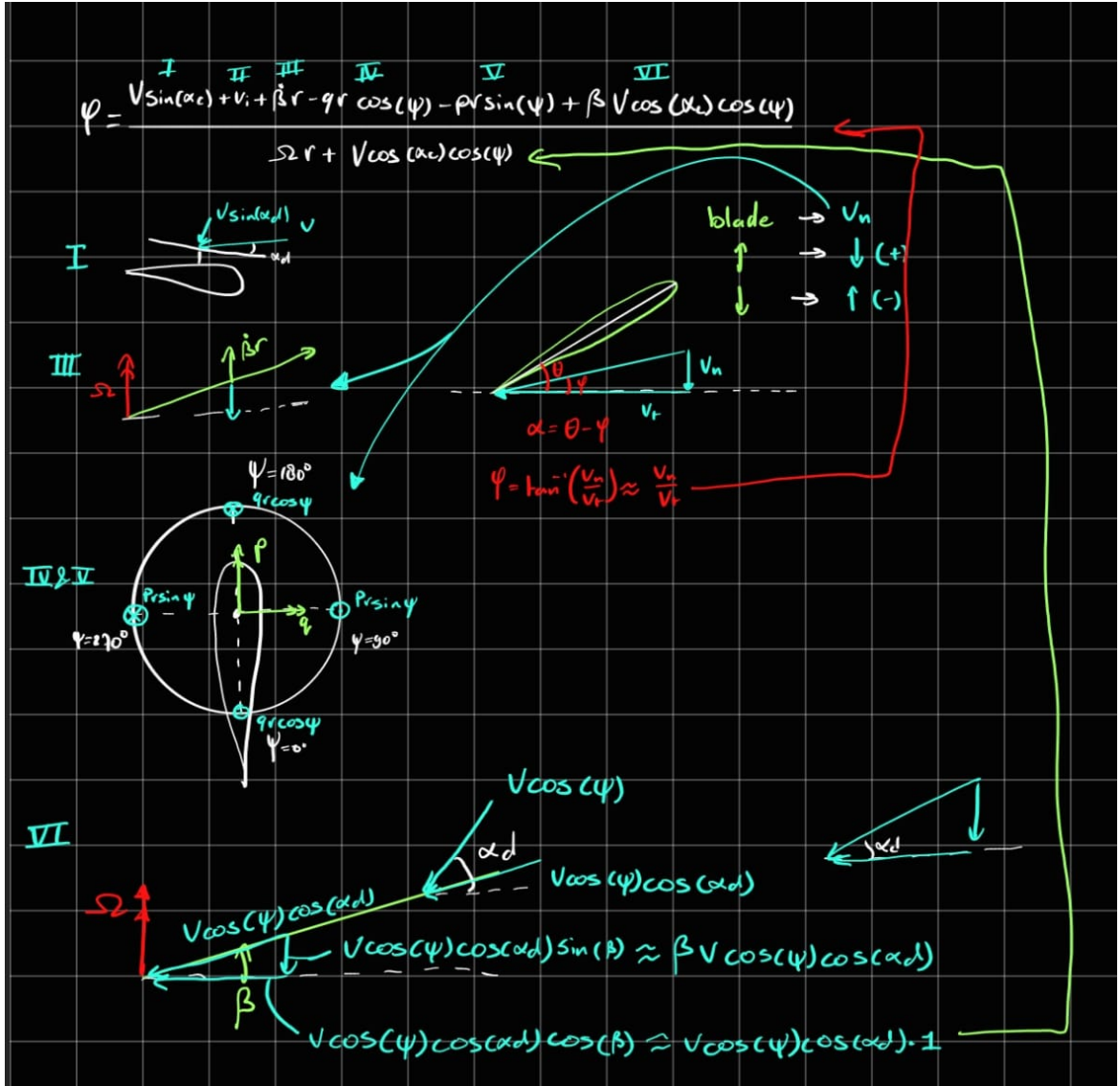


Figure 2.1: Visual representation of the inflow angle ϕ , the velocity components V_n , V_r , and the blade pitch angle θ . The figure illustrates the geometric relation underlying the expression for α in Equation 2.4.

θ in turn, is dependent on the collective angle (θ_0), the longitudinal cyclic angle (θ_1) and the lateral cyclic angle (θ_2). Since the rotor cyclic input experiences a phase lag of 90 degrees, the longitudinal and lateral components are affected by $\sin(\Psi)$ and $\cos(\Psi)$ respectively. The expression for θ thus becomes:

$$\theta = \theta_0 + \theta_1 \sin(\Psi) - \theta_2 \cos(\Psi)$$

In lecture 10, the expression for $\frac{M_a}{I_{bl}}$ is also given, this however does not yet include the contribution from the lateral-, longitudinal pitch and the roll rate. Since the equation for α is linear, these terms can simply be added. The additional terms in $\frac{M_a}{I_{bl}}$ are given by

Equation 2.6 through Equation 2.8. *Note:*

$$\mu = \frac{V \cos(\alpha_c)}{\Omega R} \quad ; \quad \lambda_c = \frac{V \sin(\alpha_c)}{\Omega R} \quad ; \quad \lambda_i = \frac{v_i}{\Omega R} \quad (2.5)$$

$$\begin{aligned} \left(\frac{M_a}{I_{bl}} \right)_p &= \frac{1}{I_{bl}} \int_0^R C_{l_\alpha} \frac{pr \sin(\Psi)}{\Omega r + V \cos(\alpha_c) \cos(\Psi)} \frac{1}{2} \rho (\Omega r + V \cos(\alpha_c) \cos(\Psi))^2 c r dr \\ &= \frac{1}{2} \frac{\rho C_{l_\alpha} c}{I_{bl}} \int_0^R p r^2 \sin(\Psi) (\Omega r + V \cos(\alpha_c) \cos(\Psi)) dr \\ &= \frac{1}{2} \frac{\rho C_{l_\alpha} c}{I_{bl}} \int_0^R (p r^3 \Omega \sin(\Psi) + p r^2 V \cos(\alpha_c) \cos(\Psi) \sin(\Psi)) dr \\ &= \frac{1}{2} \frac{\rho C_{l_\alpha} c}{I_{bl}} \left[p \frac{r^4}{4} \Omega \sin(\Psi) + p \frac{r^3}{3} V \cos(\alpha_c) \cos(\Psi) \sin(\Psi) \right]_0^R \\ &= \frac{1}{2} \frac{\rho C_{l_\alpha} c}{I_{bl}} \left[p \frac{R^4}{4} \Omega \sin(\Psi) + p \frac{R^3}{3} \Omega R \mu \cos(\Psi) \sin(\Psi) \right]_0^R \\ &= \gamma \left[\frac{1}{8} p \Omega \sin(\Psi) + \frac{1}{12} p \Omega \mu - \frac{1}{12} p \Omega \mu \cos(2\Psi) \right] \\ &= \frac{\gamma}{8} \Omega^2 \frac{p}{\Omega} \sin(\Psi) + \frac{\gamma}{12} \Omega^2 \frac{p}{\Omega} \mu - \frac{\gamma}{12} \Omega^2 \frac{p}{\Omega} \mu \cos(2\Psi) \end{aligned} \quad (2.6)$$

$$\begin{aligned} \left(\frac{M_a}{I_{bl}} \right)_{\theta_1} &= \frac{1}{I_{bl}} \int_0^R C_{l_\alpha} \theta_1 \sin(\Psi) \frac{1}{2} \rho (\Omega r + V \cos(\alpha_c) \cos(\Psi))^2 c r dr \\ &= \frac{1}{2} \frac{\rho C_{l_\alpha} c R^4}{I_{bl}} \theta_1 \sin(\Psi) \left[\frac{1}{4} + \frac{2}{3} \mu \sin(\Psi) + \frac{1}{2} \mu^2 \sin^2(\Psi) \right] \\ &= \gamma \Omega^2 \theta_1 \left[\frac{1}{6} \mu + \frac{1}{8} (1 + 3\mu^2) \sin(\Psi) - \frac{1}{6} \mu \cos(2\Psi) - \frac{1}{16} \mu^2 \sin(3\Psi) \right] \end{aligned} \quad (2.7)$$

$$\begin{aligned} \left(\frac{M_a}{I_{bl}} \right)_{\theta_2} &= \frac{1}{I_{bl}} \int_0^R C_{l_\alpha} \cdot -\theta_2 \cos(\Psi) \frac{1}{2} \rho (\Omega r + V \cos(\alpha_c) \cos(\Psi))^2 c r dr \\ &= -\frac{1}{2} \frac{\rho C_{l_\alpha} c R^4}{I_{bl}} \theta_2 \cos(\Psi) \left[\frac{1}{4} + \frac{2}{3} \mu \sin(\Psi) + \frac{1}{2} \mu^2 \sin^2(\Psi) \right] \\ &= \gamma \Omega^2 \theta_2 \left[-\frac{1}{8} (1 + \frac{1}{2} \mu^2) \cos(\Psi) - \frac{1}{6} \mu \sin(2\Psi) + \frac{1}{16} \mu^2 \cos(3\Psi) \right] \end{aligned} \quad (2.8)$$

These terms are added to the RHS of the given equation in Lecture 10. After performing the Fourier series transformation: $\beta = a_0 - a_1 \cos(\Psi) - b_1 \sin(\Psi)$, The LHS can be equated to the RHS. This means that the all the terms containing 1, $\sin(\Psi)$, $\cos(\Psi)$ can be equated thus achieving the following equalities:

$$\begin{aligned}
\left(\frac{\gamma}{12}\Omega^2\mu a_1 + \Omega^2 a_0 - \frac{\gamma}{12}\Omega^2\mu a_1\right) \cdot 1 &= \left(\frac{\gamma}{12}p\mu\Omega + \frac{\gamma}{8}\Omega^2\theta_0(1 + \mu^2) - \frac{\gamma}{6}\Omega^2(\lambda_c + \lambda_i) + \frac{\gamma}{6}\Omega^2\mu\theta_1\right) \cdot 1 \\
\Omega^2 a_0 &= \frac{\gamma}{8}\Omega^2 \left[\theta_0(1 + \mu^2) - \frac{4}{3}(\lambda_c + \lambda_i) + \frac{2}{3}\mu\frac{p}{\Omega} + \frac{4}{3}\mu\theta_1\right] \\
a_0 &= \frac{\gamma}{8} \left[\theta_0(1 + \mu^2) - \frac{4}{3}(\lambda_c + \lambda_i) + \frac{2}{3}\mu\frac{p}{\Omega} + \frac{4}{3}\mu\theta_1\right]
\end{aligned} \tag{2.9}$$

$$\begin{aligned}
\left(\frac{\gamma}{8}\Omega^2 a_1 - \frac{\gamma}{16}\Omega^2\mu^2 a_1\right) \cdot \sin(\Psi) &= \left(-2q\Omega - \frac{\gamma}{8}p\Omega + \frac{\gamma}{3}\Omega^2\mu\theta_0 - \frac{\gamma}{4}\Omega^2\mu(\lambda_c + \lambda_i) + \frac{\gamma}{8}\Omega^2\theta_1(1 + 3\mu^2)\right) \cdot \sin(\Psi) \\
\frac{\gamma}{8}\Omega^2(1 - \frac{1}{2}\mu^2)a_1 &= \frac{\gamma}{8}\Omega^2 \left[-\frac{16}{\gamma}\frac{q}{\Omega} - \frac{p}{\Omega} + \frac{8}{3}\mu\theta_0 - 2\mu(\lambda_c + \lambda_i) + \theta_1(1 + 3\mu^2)\right] \\
a_1 &= \frac{-\frac{16}{\gamma}\frac{q}{\Omega} - \frac{p}{\Omega} + \frac{8}{3}\mu\theta_0 - 2\mu(\lambda_c + \lambda_i) + \theta_1(1 + 3\mu^2)}{1 - \frac{1}{2}\mu^2}
\end{aligned} \tag{2.10}$$

$$\begin{aligned}
\left(-\frac{\gamma}{8}\Omega^2 b_1(1 + \frac{1}{2}\mu^2) + \frac{\gamma}{6}\Omega^2\mu a_0\right) \cdot \cos(\Psi) &= \left(2p\Omega - \frac{\gamma}{12}p\mu\Omega + \frac{\gamma}{8}q\Omega - \frac{\gamma}{8}\Omega^2\theta_2(1 + \frac{1}{2}\mu^2)\right) \cdot \cos(\Psi) \\
\frac{\gamma}{8}\Omega^2 \cdot -(1 + \frac{1}{2}\mu^2) &= \frac{\gamma}{8}\Omega^2 \left[\frac{16}{\gamma}\frac{p}{\Omega} - \frac{2}{3}\frac{p}{\Omega}\mu + \frac{q}{\Omega} - \frac{4}{3}\mu a_0 + \theta_2(1 + \frac{1}{2}\mu^2)\right] \\
b_1 &= \frac{-\frac{16}{\gamma}\frac{p}{\Omega} + \frac{2}{3}\frac{p}{\Omega}\mu - \frac{q}{\Omega} + \frac{4}{3}\mu a_0}{1 + \frac{1}{2}\mu^2} - \theta_2
\end{aligned} \tag{2.11}$$

2.2 Blade flapping motion

Now that the equations for the coning angles: a_0 , a_1 and b_1 are known, the flapping motion for this helicopter can be solved. Relevant helicopter and flight parameters are tabulated in Table 2.1.

Firstly, the induced velocity was calculated. This was done in the same manner as described in section 1.2 and yielded a value of 6.48 m/s for the forward velocity of 20 m/s. With this value known, λ_i could be calculated according to Equation 2.5. Next, μ and λ_c had to be determined to calculate the coning parameters. These are, however, a value of α_c which is yet unknown. Based on the definition of the coning parameter a_1 , it can be observed that $\alpha_c = \alpha_d - a_1$ with α_d easily determinable by Equation 1.14. The value for a_1 could therefore be directly solved numerically by entering all known parameters and coding μ and λ_c as a function of a_1 . This directly returned the value for a_1 according to Equation 2.10. With the value for a_1 now known, μ and λ_c could easily be calculated by Equation 2.5. Then filling these known parameters into Equation 2.9 and Equation 2.11 then returned

$$1 \frac{RPM}{60} \cdot 2\pi$$

²The calculation using the data from the helicopter & assuming $C_{l_\alpha} = 0.9 \cdot 2\pi \text{ rad}^{-1}$ and $I_{bl} = \frac{1}{3}mR^2$ gave a Lock number of around 5 which was much lower than typical numbers. It was therefore chosen to take the Lock number to be 9.

Parameter	Value	Unit
Helicopter parameters		
R	5.2	m
mass	2910	kg
Ω	41.36	rad/s
$\sum(C_D S)_S$	2.00	m^2
γ^2	9	rad ⁻¹
Flight parameters		
V	20	m/s
q	20	deg/s
p	10	deg/s
θ_0	6	deg
θ_1	2	deg
θ_2	1	deg
ρ	1.225	kg/m ³

Table 2.1: Parameters used for the blade flapping calculation.

the values for a_0 and b_1 after which the flapping angle as a function of azimuth could be determined by Equation 2.12 with the numerically achieved coning parameters tabulated in Table 2.2. The flapping angle (in degrees) is plotted against the Azimuth angle (in degrees) over one full rotation of the rotor blade in Figure 2.2.

$$\beta = a_0 - a_1 \cos(\Psi) - b_1 \sin(\Psi) \quad (2.12)$$

a_0	0.0816 rad
a_1	0.0375 rad
b_1	-0.0157 rad

Table 2.2: Coning parameters from the numerical solution of the fourier transformed flapping EOM.

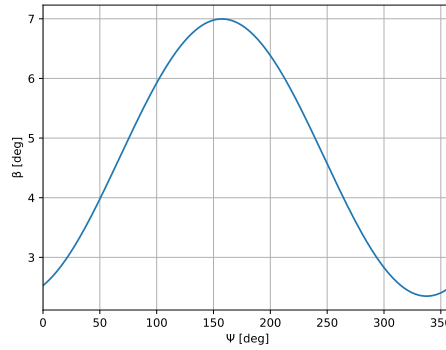


Figure 2.2: flapping angle β plotted against azimuth angle Ψ

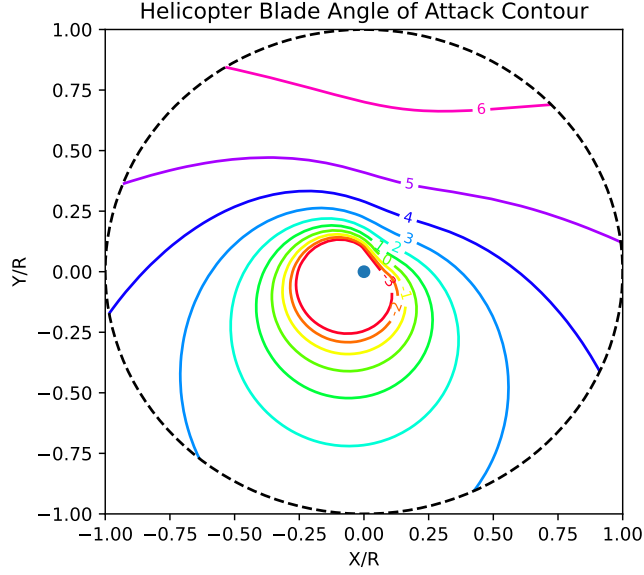


Figure 2.3: Lines of constant local angle of attack over the rotor with $\Psi = 180^\circ$ at the top of the figure.

2.3 Angle of attack of the rotor

With the flapping angle of the rotor known for a given azimuth angle, so too can the angular velocity $\dot{\beta}$ be determined according to Equation 2.13.

$$\dot{\beta} = \frac{d\beta}{d\Psi} \frac{d\Psi}{dt} = (a_1 \sin(\Psi) - b_1 \cos(\Psi)) \cdot \Omega \quad (2.13)$$

With all parameters known, the local blade angle of attack can be calculated according to Equation 2.4 which is now only a function of the azimuth Ψ and the radial position on the blade r . These local angles of attack are then compiled in a contour plot which displays contours of constant angle of attack over the normalized x and y coordinates of the rotor. This contour plot is displayed in Figure 2.3.

Here, it is important to note that for the analysis, the inner 0.5m of the rotor has been omitted from the calculations as the small radius meant that the radial velocity is extremely small in this region. This combined with the induced velocity being assumed constant according to the actuator disc theory means that the ratio between the normal and tangential velocity components would grow unnaturally large thus resulting in extremely high negative angles of attack. As this behavior is not relevant to the overall analysis of the angle of attack over the rotor, this region has thus been omitted from the plot.

2.4 Effect of twist on the coning angle in hover.

In hover, the expression for M_a/I_{bl} becomes:

$$\begin{aligned}
\frac{M_a}{I_{bl}} &= \frac{1}{I_{bl}} \int_0^R C_{l_\alpha} \left(\theta_0 - \frac{r}{R} \theta_{tw} - \frac{v_i + \dot{\beta} r}{\Omega r} \right) \frac{1}{2} \rho (\Omega r)^2 c r dr \\
&= \frac{1}{2} \frac{\rho C_{l_\alpha} c}{I_{bl}} \int_0^R \left(\theta_0 \Omega^2 r^3 - \frac{\Omega^2}{R} \theta_{tw} r^4 - v_i \Omega r^2 - \dot{\beta} \Omega r^3 \right) dr \\
&= \frac{1}{2} \frac{\rho C_{l_\alpha} c}{I_{bl}} \left[\frac{1}{4} \theta_0 \Omega^2 R^4 - \frac{1}{5} \theta_{tw} \Omega^2 R^4 - \frac{1}{3} \Omega v_i R^3 - \frac{1}{4} \Omega \dot{\beta} R^4 \right] \\
&= \frac{\gamma}{8} \Omega^2 \left[\theta_0 - \frac{4}{5} \theta_{tw} - \frac{4}{3} \lambda_i - \frac{\dot{\beta}}{\Omega} \right]
\end{aligned}$$

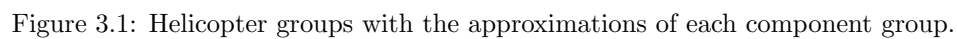
This results in the definition for a_0 coming to:

$$a_0 = \frac{\gamma}{8} \left[\theta_0 - \frac{4}{3} \lambda_i \right] - \frac{\gamma}{10} \theta_{tw}$$

It can thus be seen that by introducing a linear twist $\frac{r}{R} \theta_{tw}$, the coning angle a_0 in hover is reduced by $\frac{\gamma}{10} \theta_{tw}$ rad.

Part 3 - Helicopter Flight Mechanics Manoeuvre Simulation

In order to estimate the moment of inertia about the y-axis (I_y) of the helicopter, first some assumptions had to be made. Firstly, the side view drawing of the EC135 as provided by Eurocopter [4] was imported into adobe illustrator which allowed for the positional analysis of certain components. The centre of gravity was then estimated to be directly underneath the rotor hub above the centre of the body as the rotor assembly and engines make up a sizeable portion of the helicopter mass. The helicopter was then divided into four main components: The rotor, the main body, the tail and the engines. Typical mass fractions for a helicopter were used [8] and grouped into these mentioned groups. The summary of this can be found in Table 3.1. The moment of inertia about the centre of mass can then be estimated for each of these four groups. To do this, the rotor group was estimated as a disc, the main body as an ellipsoid and the tail & engine group as a cylinder. This can be seen in Figure 3.1.



Component	mass fraction	mass [kg]
Rotor group		
Rotors	0.101	146.96
Total	0.101	146.96
Main body group		
Fuselage	0.143	208.0
Furnishings	0.118	171.7
Avionics	0.079	114.95
Electronics	0.079	114.95
Landing gear	0.073	106.21
Flight controls	0.032	46.56
Instruments	0.03	43.65
Auxiliary power system	0.026	37.83
Total	0.58	843.9
Tail group		
Tail	0.07	101.85
Total	0.07	101.85
Engine group		
Engines	0.123	178.97
Drive system	0.071	103.3
Fuel system	0.034	49.74
Nacelle	0.021	30.56
Total	0.249	362.295

Table 3.1: Helicopter component group mass make-up.

The moment of inertia for each of the groups can then be calculated:

Rotor group

The rotor is approximated as a disc at a vertical distance of $r = 1.11$ m from the c.g. (from the illustration software.):

$$\begin{aligned}
 (I_y)_{Rotor} &= \frac{1}{4} m_{rotor} R^2 + m_{Rotor} r^2 \\
 &= \frac{1}{4} \cdot 146.96 \cdot 5.2^2 + 146.96 \cdot 1.11^2 = 1150 [kgm^2]
 \end{aligned} \tag{3.1}$$

Main body group

The main body is approximated as a solid ellipsoid. This is assumed valid as the internal components of the main body make up a sizeable portion of the main body total mass. The ellipsoid has a height of $h = 1.1$ m and a length of $l = 2.9$ m with the centroid located about 0.7 m away from the c.g.:

$$\begin{aligned}
 (I_y)_{body} &= \frac{m_{body} h^2}{5} \left(1 + \frac{l^2}{h^2} \right) + m_{body} r^2 \\
 &= \frac{843.9 \cdot 1.1^2}{5} \left(1 + \frac{2.9^2}{1.1^2} \right) + 843.9 \cdot 0.7^2 = 2057 [kgm^2]
 \end{aligned} \tag{3.2}$$

Tail group

The tail was estimated as a cylinder of length $L = 2.3$ m and radius $R = 0.345$ m. Its centroid is at a distance $r = 4.4$ m:

$$\begin{aligned}(I_y)_{tail} &= \frac{1}{4}m_{tail}R^2 + \frac{1}{12}m_{tail}L^2 + m_{tail}r^2 \\ &= \frac{1}{4}101.85 \cdot 0.345^2 + \frac{1}{12}101.85 \cdot 2.3^2 + 101.85 \cdot 4.4^2 = 2003[kgm^2]\end{aligned}\tag{3.3}$$

Engine group

Finally, the engine was estimated as a cylinder of length $L = 1.4$ m and a radius $R = 0.275$ m with its centroid at a distance $r = 0.5$ m from the c.g.:

$$\begin{aligned}(I_y)_{engine} &= \frac{1}{4}m_{engine}R^2 + \frac{1}{12}m_{engine}L^2 + m_{engine}r^2 \\ &= \frac{1}{4}362.295 \cdot 0.275^2 + \frac{1}{12}362.295 \cdot 1.4^2 + 362.295 \cdot 0.5^2 = 158[kgm^2]\end{aligned}\tag{3.4}$$

Total moment of inertia

The total moment of inertia I_y can thus be determined to be:

$$\begin{aligned}I_{y,tot} &= I_{y,rotor} + I_{y,body} + I_{y,tail} + I_{y,engine} \\ &= 1150 + 2057 + 2003 + 158 = 5370[kgm^2]\end{aligned}$$

3.2 Trim Calculation

To compute the trim condition of the helicopter in (forward), stationary, horizontal flight, the 3 degree of freedom longitudinal model is used. The control inputs (collective pitch θ_o and longitudinal cyclic pitch θ_c) as a function of forward velocity V are obtained, ranging from hover to the helicopter's maximum forward speed.

The methodology is based on the theoretical formulation in Chapter 13 (p.105) of the course book, the computation scheme from Chapter 11.7 (p.95), and the slides of Lecture 11. The numerical implementation uses Python's `fsolve` from the `scipy.optimize` library to solve the system.

Trim Assumptions and Force Balance

In order to be in trim, the helicopter must have no accelerations, i.e.:

$$\dot{u} = \dot{w} = \dot{q} = \dot{\theta}_f = 0$$

and the surge and heave follow from the constant body pitch angle θ_f :

$$u = V \cos \theta_f \tag{3.5}$$

$$w = V \sin \theta_f \tag{3.6}$$

The equilibrium conditions from the force balance equations are (from Eqs. 110–112 in the book):

$$0 = -W \sin \theta_f - D \cos \theta_f + T \sin(\theta_c - a_1) \quad (3.7)$$

$$0 = W \cos \theta_f - D \sin \theta_f - T \cos(\theta_c - a_1) \quad (3.8)$$

$$0 = -Th \sin(\theta_c - a_1) \quad (3.9)$$

From Equation 3.9 it follows that:

$$\theta_c = a_1 \quad (3.10)$$

This implies the tip-path plane is parallel to the X-axis and the thrust vector intersects the center of gravity.

Substituting $\theta_c = a_1$ into the first equation gives the fuselage pitch angle:

$$\tan \theta_f = -\frac{D}{W} \quad (3.11)$$

and using the second equation, the required thrust then becomes:

$$T = \sqrt{W^2 + D^2} \quad (3.12)$$

Critically, using the definition of u, w in trim, $\alpha_c = \theta_c - \theta_f$. Together with the fact that $\theta_c = a_1$, that angle of attack of the rotor disk in trim is as follows:

$$\alpha_d = \theta_o - \theta_f = a_1 - \theta_f = \arctan\left(\frac{D}{W}\right) \approx \frac{D}{W} \quad (3.13)$$

Assuming small angles.

According to the Glauert method, the nondimensional induced velocity λ_i satisfies:

$$C_T = 2\lambda_i \sqrt{\left(\frac{V}{\Omega R} \cos \alpha_d\right)^2 + \left(\frac{V}{\Omega R} \sin \alpha_d + \lambda_i\right)^2} \quad (3.14)$$

Also, from the flapping equation:

$$a_1 = \frac{\frac{8}{3}\mu\theta_0 - 2\mu(\lambda_c + \lambda_i)}{1 - \frac{1}{2}\mu^2} (= \theta_c) \quad (3.15)$$

Since all angles are considered to be small, we define the advance ratio as:

$$\mu = \frac{V}{\Omega R} \quad (3.16)$$

and the climb inflow component as:

$$\lambda_c = \mu(\alpha_d + \theta_c) \quad (3.17)$$

Obtaining the Trim Conditions

To solve for θ_c and θ_o , we rewrite Equation 3.15 and Equation 3.14, to the following:

$$f = C_T - \frac{c_{l\alpha} \cdot \sigma}{4} \left[\frac{2}{3} \theta_o \left(1 + \frac{3}{2} \mu^2 \right) - (\lambda_i + \lambda_c) \right] \quad (3.18)$$

$$g = \theta_c - \frac{\frac{8}{3} \mu \theta_o - 2\mu(\lambda_i + \lambda_c)}{1 - \frac{1}{2} \mu^2} \quad (3.19)$$

The system is solved using `fsolve`, which chooses θ_o and θ_c such that $f, g \rightarrow 0$.

Implementation Overview

The code iteratively solves for trim conditions over a range of forward velocities. For each velocity:

- Fuselage drag D is computed using $D = \frac{1}{2} \rho V^2 C_{dS}$.
- θ_f is computed via $\tan \theta_f = -D/W$.
- Required thrust coefficient C_T is derived.
- The normalized induced λ_i is obtained numerically, using the loop suggested on p.95 of the book and `fsolve`.
- The pitch inputs θ_c and θ_o are found by solving the trim equations.

Results

Figure 3.2 shows the computed variation of collective and cyclic pitch inputs as a function of forward velocity. These curves represent how a pilot would need to adjust controls to maintain steady-level flight.

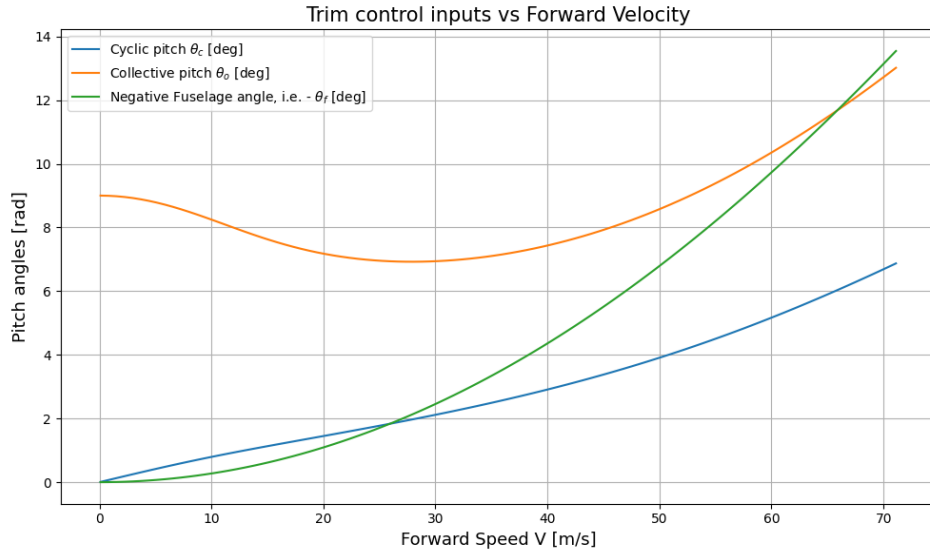


Figure 3.2: Trim control inputs θ_c (cyclic) and θ_o (collective) as functions of forward velocity V . As well as the negative fuselage trim angle θ_f , showing a pitch nose down angle.

The implementation accurately reproduces the theoretical trim behavior presented in Chapter 13, specifically Figure 13.1. Serving as a validation to our result. Now, as will be used later on, since these are equilibrium position, each trim speed corresponds to one value for the pitch angles, induced velocity and fuselage angle. The most important trim speeds, which will be used for the ADS33 manoeuvre, are shown in the table below:

Table 3.2: Trim values at different airspeeds

Speed (knots)	V (m/s)	θ_0 (rad)	θ_c (rad)	θ_f (rad)	v_i (m/s)
70	35.96	0.1249	0.0448	-0.0615	3.4030
90	46.01	0.1404	0.0608	-0.1005	2.6759
110	56.77	0.1695	0.0826	-0.1523	2.1851

3.3 Manoeuvre Simulation (Q3.2)

In this section we will be simulating the three degree of freedom **speed control** manoeuvre described in the US Army Aeronautical Design Standard ADS-33. The goal is to evaluate the helicopter's ability to transition between different airspeeds while maintaining trim and acceptable flight qualities, as required in mission-criteria.

Model Description and Equations of Motion

The nonlinear simulation is based on the equations from Chapter 11.7 (p. 95) of the course book. The state and control vectors are defined as:

$$\mathbf{x} = \begin{bmatrix} u & w & q & \theta_f & v_i \end{bmatrix}^\top \quad \text{and} \quad \mathbf{u}_c = \begin{bmatrix} \theta_o & \theta_c \end{bmatrix}^\top$$

where v_i has been added to avoid the computationally heavy derivation of λ_i , as suggested in Appendix A of the assignment description. The corresponding equations of motion are:

$$\dot{u} = -g \sin \theta_f - \frac{D_{\text{fus}}}{W} \cdot \frac{u}{V} + \frac{T}{W} \sin(\theta_c - a_1) - qw \quad (3.20)$$

$$\dot{w} = g \cos \theta_f - \frac{D_{\text{fus}}}{W} \cdot \frac{w}{V} - \frac{T}{W} \cos(\theta_c - a_1) + qu \quad (3.21)$$

$$\dot{q} = -\frac{T}{I_y} \cdot h \sin(\theta_c - a_1) \quad (3.22)$$

$$\dot{\theta}_f = q \quad (3.23)$$

$$\dot{v}_i = \frac{1}{\tau} (C_{T,\text{elem}} - C_{T,\text{Glauert}}) \quad (3.24)$$

where a_1 is the flapping angle and $\tau = 0.2$ s is a time constant.

These equations are implemented in the Python function `heli_dynamics`, which defines the full nonlinear dynamics of the system. This function relies on `heli_forces` to compute the aerodynamic forces required in the equations above.

The `heli_forces` function calculates the following quantities:

- Rotor thrust T ,
- Fuselage drag D_{fus} ,

- Flapping angle a_1 ,
- Blade element thrust coefficient $C_{T,\text{elem}}$,
- Glauert momentum theory thrust coefficient $C_{T,\text{Glauert}}$.

This is done by first computing the airspeed $V = \sqrt{u^2 + w^2}$ and determining the control plane angle of attack:

$$\alpha_c = \theta_c - \arctan\left(\frac{w}{u}\right)$$

with a quadrant correction applied when $u < 0$, i.e $\alpha_c = \alpha_c + \pi$. Using this, the advance ratio and inflow components are computed:

$$\mu = \frac{V}{\Omega R} \cos \alpha_c, \quad \lambda_i = \frac{v_i}{\Omega R}, \quad \lambda_c = \frac{V \sin \alpha_c}{\Omega R}$$

The flapping angle is then given by:

$$a_1 = \frac{\frac{8}{3}\mu\theta_o - 2\mu(\lambda_i + \lambda_c) - \frac{16}{\gamma\Omega}q}{1 - \frac{1}{2}\mu^2}$$

The thrust coefficients are calculated as:

$$C_{T,\text{elem}} = \frac{1}{4}c_{l\alpha} \cdot \sigma \left[\frac{2}{3}\theta_o \left(1 + \frac{3}{2}\mu^2 \right) - (\lambda_i + \lambda_c) \right] \quad (3.25)$$

$$C_{T,\text{Glauert}} = 4a_1 \sqrt{\left(\frac{V \cos(\alpha_c - a_1)}{\Omega R} \right)^2 + \left(\frac{V \sin(\alpha_c - a_1)}{\Omega R} + \lambda_i \right)^2} \quad (3.26)$$

Finally, the thrust and fuselage drag are given by:

$$T = C_{T,\text{elem}} \cdot \rho \Omega^2 \pi R^4, \quad D_{\text{fus}} = \frac{1}{2} \rho C_{dS} V^2$$

These outputs are returned by `heli_forces` and used within `heli_dynamics` to compute the state derivatives at each timestep of the simulation. The time integration of the system is carried out using a fixed-step explicit Euler method. At each iteration, the next state is computed using:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t \cdot \dot{\mathbf{x}}_k$$

where Δt is the simulation timestep, chosen as $\Delta t = 0.01$ s. This method commonly used since it is a simple and computationally efficient way to obtain system dynamics.

Validating Open Loop System Model (without Controller)

To validate if the model that has been generated works as intended, we will first look at the open-loop model (without a feedback controller), showing the natural unstable behaviour of the helicopter. To do so, three different inputs of the cyclic pitch are introduced to the system, generating the open-loop response. The state trajectories of the forward velocity u and fuselage angle θ_f are compared to those simulated in Ch.12 of the book. $\theta_{c,\text{trim,hover}}$ has been taken from Q3.1 for $V = 0$, i.e. hover. The three validation inputs are:

- **Case 1 – Step input (Fig. 12.1):** At $t = 0.5$ s, a step of 1 degree is applied to the cyclic pitch input θ_c for 0.5 seconds.

$$\theta_c(t) = \begin{cases} \theta_{c,\text{trim,hover}} + 1^\circ, & 0.5 < t < 1.0 \\ \theta_{c,\text{trim,hover}}, & \text{otherwise} \end{cases}$$

- **Case 2 – Proportional feedback (Fig. 12.3):** After $t = 5$ s, a proportional feedback law is applied based on the fuselage pitch angle.

$$\theta_c(t) = \begin{cases} \theta_{c,\text{trim,hover}}, & t \leq 5 \\ 0.2 \cdot \theta_f(t), & t > 5 \end{cases}$$

- **Case 3 – PD-like feedback (Fig. 12.4):** After $t = 5$ s, a feedback law is used based on both fuselage pitch angle and pitch rate:

$$\theta_c(t) = \begin{cases} \theta_{c,\text{trim,hover}}, & t \leq 5 \\ 0.2 \cdot \theta_f(t) + 0.2 \cdot q(t), & t > 5 \end{cases}$$

The response of $u(t)$ and $\theta_f(t)$ for each of these three cases is shown below. The figures confirm that the model qualitatively reproduces the characteristics of the behavior presented in the book.

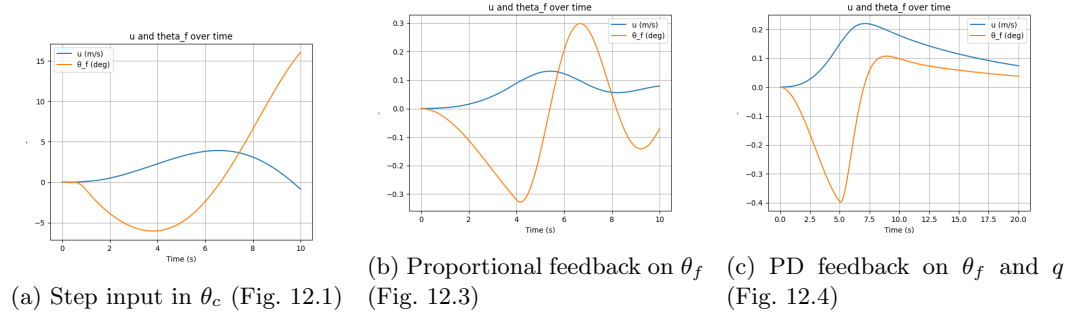


Figure 3.3: Validation of helicopter model response using three cyclic pitch control inputs. Each figure shows the evolution of forward velocity $u(t)$ and fuselage pitch angle $\theta_f(t)$.

Stepwise Implementation: an Altitude-Hold Controller

In order to progressively build our controller able to achieve the objective, we start by making an altitude-hold controller. For this, we followed the control strategy described in Appendix A.3 of the assignment. As a first step, a proportional controller is implemented to regulate vertical height h using the collective pitch θ_o . This approach provides a simple yet insightful baseline to evaluate control behavior, especially the presence of steady-state error.

The control law used in this initial step is given by:

$$\theta_o = \theta_{\text{gen}} + K_1(h_{\text{des}} - h) \quad (3.27)$$

where θ_{gen} is a nominal collective pitch (chosen as the collective pitch associated with the initial trim speed, from Q3.1), K_1 is the proportional gain, h_{des} is the desired altitude, and h is the actual altitude. The vertical velocity c , which has been defined positive upwards, is computed from the helicopter's velocity components u and w and the fuselage pitch angle θ_f , using:

$$c = u \sin(\theta_f) - w \cos(\theta_f) \quad (3.28)$$

To monitor this altitude response, we introduce this altitude h as an additional state in the simulation. However, to maintain clarity in the helicopter model structure, h is not integrated into the main `heli_dynamics` function. Instead, it is treated separately and updated manually, using Euler, at each simulation step by integrating $c = \dot{h}$ s.t. $h_{k+1} = h_k + \Delta t \cdot \dot{h}$. Applying this, with $K_1 = -0.01$, the following response can be obtained (see Figure 3.4).

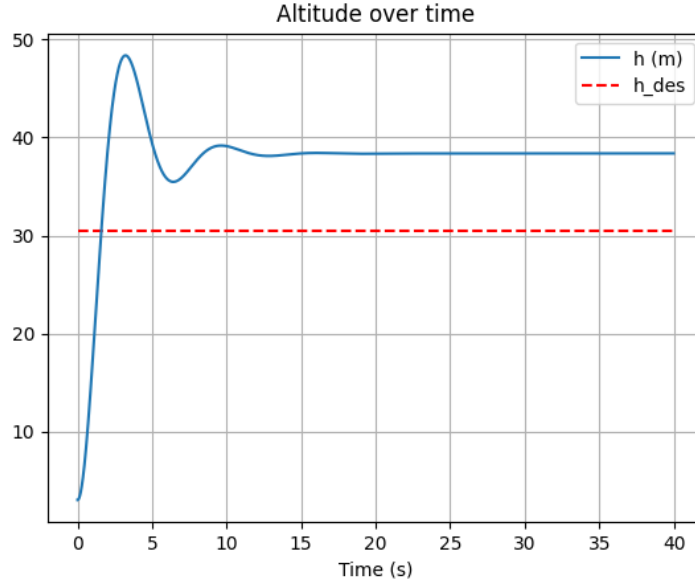


Figure 3.4: Altitude response of a proportional controller. Blue line indicates the altitude response and the red line the desired altitude (100ft)

While this proportional control improves the behavior in terms of altitude control, a static error is clearly observed. To address this steady-state error observed in the proportional controller, we extend the controller to include integral action. The resulting PI controller modifies the collective input θ_o by integrating the vertical velocity error over time:

$$\theta_o = \theta_{\text{gen}} + K_1(h_{\text{des}} - h) + K_2\Delta h \quad (3.29)$$

$$\Delta h = \int_0^t (h_{\text{des}} - h) d\tau \quad (3.30)$$

Here, K_2 is the integral gain and Δh is the accumulated altitude error. In our implementation, Δh is treated as another "state" and is updated using numerical integration via:

$$\dot{\Delta h}_k = (h_{\text{des}} - h) \Rightarrow \Delta h_{k+1} = \Delta h_k + \Delta t \cdot \dot{\Delta h}_k \quad (3.31)$$

The inclusion of integral action (with $K_2 = -0.001$) eliminates the residual offset and allows the system to stabilize around the target altitude, as can in the following figure (Figure 3.5)

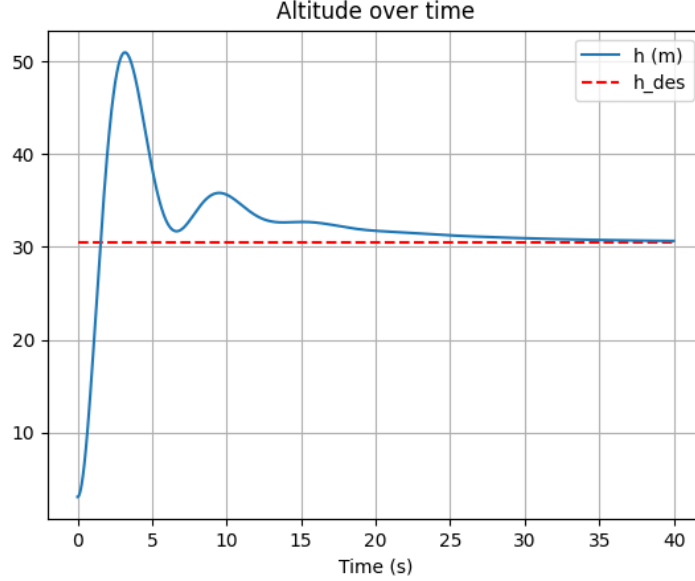


Figure 3.5: Altitude response of a proportional and integral controller. Blue line indicates the altitude response and the red line the desired altitude (100ft)

As the final step in constructing the altitude-hold controller, we replace the fixed c_{des} with a dynamic value derived from the altitude error. This introduces an outer control loop where the altitude error generates a new vertical velocity target:

$$c_{\text{des}} = K_3(h_{\text{des}} - h) \quad (3.32)$$

where K_3 is a proportional gain that maps altitude error to vertical velocity, and h_{des} is the desired altitude. This results in a cascaded control structure:

- The outer loop converts altitude error into a desired vertical velocity.
- The inner loop (PI controller) adjusts the collective pitch to track this desired velocity.

This nested approach allows the helicopter to adjust its collective pitch in order to smoothly and accurately track the target altitude. The overall control law for the collective becomes:

$$\theta_o = \theta_{\text{gen}} + K_1(K_3(h_{\text{des}} - h) - c) + K_2\Delta\theta \quad (3.33)$$

where $\dot{\Delta\theta} = K_3(h_{\text{des}} - h) - c$ and $\dot{h} = c$.

These additions to the controller were implemented step by step in the simulation code. By maintaining h and Δh outside of the main `heli_dynamics` function, we preserve a clear separation between helicopter dynamics and controller logic. This facilitates both debugging and future extensions. To conclude this step, we visualize the system’s response in terms of the body-axis velocities u and w , as well as the control inputs θ_0 and θ_c . These plots provide further insight into how the controls applied to the system behave, and their resulting dynamics.

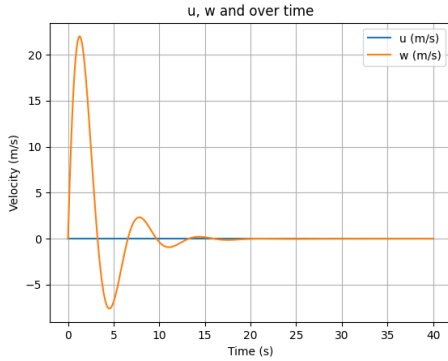


Figure 3.6: Body velocities u and w over time.

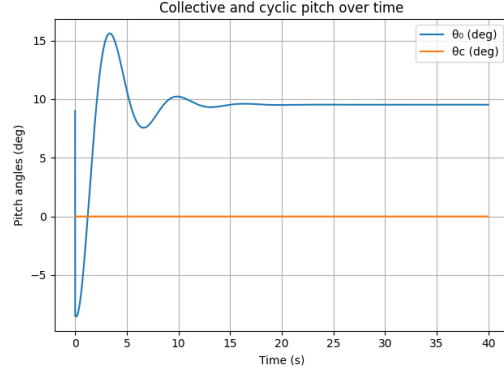


Figure 3.7: Collective (θ_0) and cyclic (θ_c) pitch angles. θ_0 is actively controlled, while θ_c remains constant.

Figure 3.6 shows that w plays a dominant role in altitude control, as expected. The horizontal component u remains near zero since horizontal motion is not commanded. Meanwhile, Figure 3.7 confirms that the control effort is achieved via collective pitch θ_0 , with the longitudinal cyclic pitch θ_c held constant throughout the simulation.

This completes the stepwise implementation of the altitude-hold controller.

Stepwise Implementation: a Fuselage Angle Controller

Having successfully implemented altitude hold, we now extend our control architecture with a controller that regulates the fuselage pitch angle θ_f . Since altitude is automatically controlled by the previous controller, we can now focus on tracking trim trajectories defined by a desired forward speed V_{ref} .

From Question 3.1, we know that for each trim speed, there exists a unique set of trim values: collective pitch θ_o , longitudinal cyclic pitch θ_c , fuselage pitch angle θ_f , and induced inflow v_i . These trim values, computed across a range of forward velocities, are stored in a data file `trim_data.npz`. The function `get.trim(V)` provides these values for any desired velocity V .

Our control goal is to manoeuvre the helicopter such that it reaches the desired trim condition for a given V_{ref} . Since there is a one-to-one mapping between V_{ref} and the corresponding fuselage pitch angle $\theta_{f,\text{ref}}$, we use θ_f as the controlled variable. This strategy allows us to “pull” the helicopter fuselage angle toward the desired pitch, and thus indirectly toward the desired speed when combined with the altitude controller.

As in the altitude controller design, we follow a stepwise approach—starting with a proportional controller.

Proportional Control. We first implement a proportional (P) controller that acts on the difference between the reference and current fuselage pitch angle:

$$\theta_c = \theta_{c_0} + K_p(\theta_{f_{\text{ref}}} - \theta_f) \quad (3.34)$$

where:

- θ_c is the control input longitudinal cyclic pitch,
- θ_{c_0} is the trim cyclic pitch at the initial trim speed from Q3.1,
- $\theta_{f_{\text{ref}}}$ is the reference fuselage pitch angle (from trim),
- θ_f is the current fuselage pitch angle,
- K_p is the proportional gain.

This controller intuitively attempts to align the fuselage with the reference orientation by applying corrective cyclic pitch. As shown in Figure 3.8, the controller is able to pull the fuselage toward the target pitch angle, but a steady-state error remains.

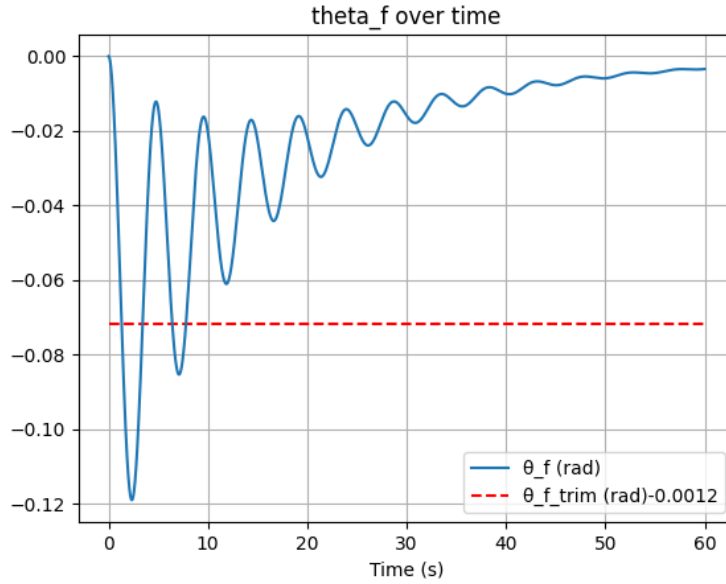


Figure 3.8: Fuselage angle response using a proportional controller. A steady-state error remains due to the lack of integral action.

PI Control. To address the steady-state error observed with the P controller, we added an integral term to form a PI controller. This resulted in the following control law:

$$\theta_c = \theta_{c_0} + K_p(\theta_{f_{\text{ref}}} - \theta_f) + K_i \int_0^t (\theta_{f_{\text{ref}}} - \theta_f) d\tau \quad (3.35)$$

Like has been done in attitude this is treated as a "state" and summed using Euler integration. The integral term enables the controller to eliminate persistent errors by accumulating deviations over time. Figure 3.9 shows the fuselage pitch angle response under this PI control.

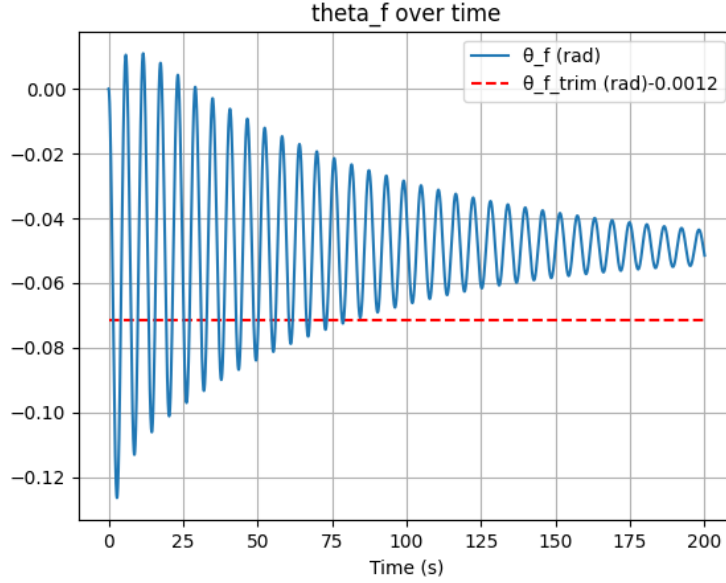


Figure 3.9: Fuselage pitch response using a PI controller. Oscillations are damped over time, but convergence to trim is slow.

Compared to the P controller, the steady-state error is slightly reduced. However, the figure also reveals a key limitation: the system still exhibits large oscillations that take a long time to decay. This behavior stems from a lack of damping in the system, which is not addressed by proportional or integral action alone.

Derivative Action Using Pitch Rate. To overcome this issue, we extend the controller with a derivative term. The derivative action reacts to the rate of change of the error, effectively adding damping to the system and suppressing oscillatory behavior. This will result in a full PID controller. In our case, the most natural derivative of the fuselage angle θ_f is the pitch rate q , as they are directly related by:

$$q = \dot{\theta}_f \quad (3.36)$$

Rather than computing error term numerically, we use q directly in the control law:

$$\theta_c = \theta_{c0} + K_p(\theta_{f\text{ref}} - \theta_f) + K_i \int_0^t (\theta_{f\text{ref}} - \theta_f) d\tau + K_d q \quad (3.37)$$

This approach has two key benefits: (1) q is already available as a system state, and (2) it aligns perfectly with the physical meaning of derivative action on fuselage pitch.

Figure 3.10 shows the time response of θ_f and its corresponding pitch rate q . Note how q is approximately 90° out of phase with θ_f , which is typical in underdamped oscillatory systems. This phase relationship indicates that q reaches its peak when θ_f is crossing its mean value, making it an ideal candidate for damping the oscillations through negative feedback.

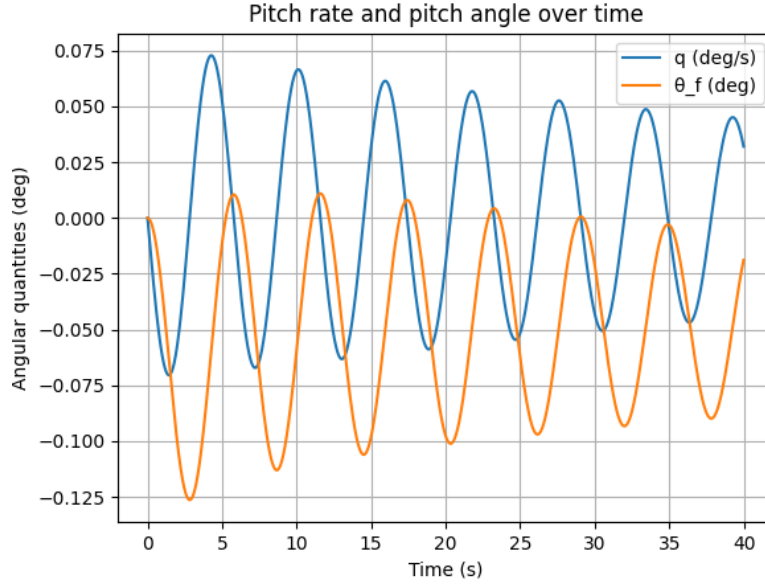


Figure 3.10: Pitch rate q and fuselage angle θ_f over time. The signals are out of phase, confirming that q is a suitable damping input.

After tuning, the gains have been found as $K_p = 0.4$, $K_i = 0.8$, and $K_d = -0.8$. The system achieves a well-damped response with minimal overshoot and convergence to the trim value. The derivative action, using q , effectively suppresses the oscillations.

Figure 3.11 illustrates the fuselage pitch angle response under the PID controller. The oscillations are significantly reduced, and the trim angle is reached within a reasonably short time.

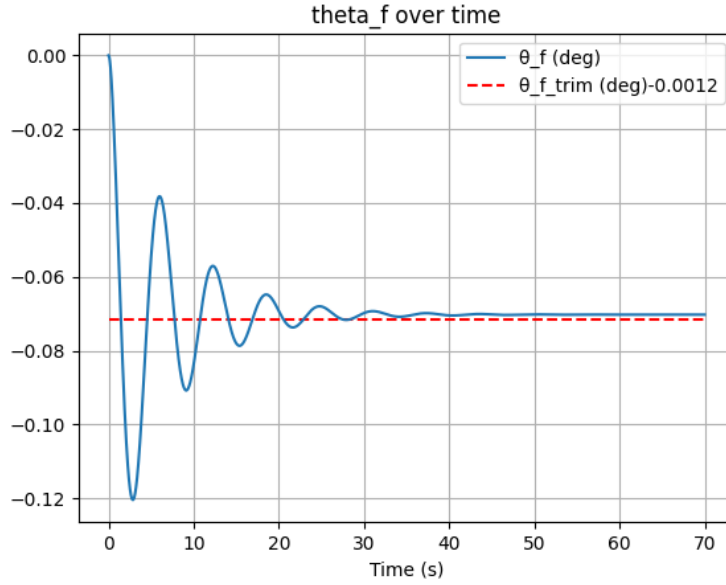


Figure 3.11: Fuselage pitch response using a tuned PID controller. The system reaches trim efficiently with minimal overshoot and no steady-state error.

With this controller architecture in place, the helicopter is now capable of smoothly following fuselage angle trajectories associated with a desired forward velocity V_{ref} , as required in the ADS-33 speed control manoeuvre.

Results

In this section, we evaluate whether the designed controller meets the objectives defined in the ADS-33 "speed control" manoeuvre. This manoeuvre consists of a sequence of speed transitions: starting at 90 knots, decelerating to 70 knots, accelerating back to 90 knots, and finally accelerating to 110 knots. During this entire manoeuvre, the helicopter must also maintain its altitude within 100ft. The time needed from changing from one trim airspeed to another should be within 1 minute, 2 minutes is adequate. Trim hands-off at target airspeed within 3 knots, where 5 knots is adequate. Lastly, heading should be maintained within 5 degrees, where 10 degrees is adequate.

Altitude Tracking. As shown in Figure 3.12, the altitude tracking is highly successful. The helicopter remains within a narrow band well within the 100 ft reference throughout the entire manoeuvre. The deviations are minimal and quickly damped, which confirms that the nested altitude controller with PI compensation performs robustly under these conditions.

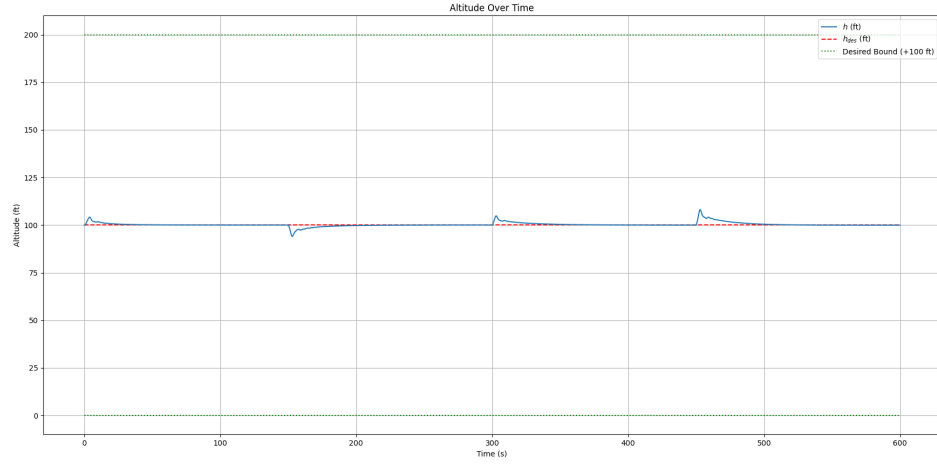


Figure 3.12: Altitude tracking during the ADS-33 speed profile. The controller maintains altitude within tight bounds.

Pitch and Control Inputs. The control inputs and pitch behavior are shown in Figures 3.13 and 3.14. As expected, significant cyclic input is required during speed transitions to pitch the fuselage up or down. These transitions result in corresponding variations in θ_f and q . Notably, θ_c shows sharp but controlled variations with some oscillation, while θ_0 shifts in steps to support the changing lift demands due to the velocity transitions.

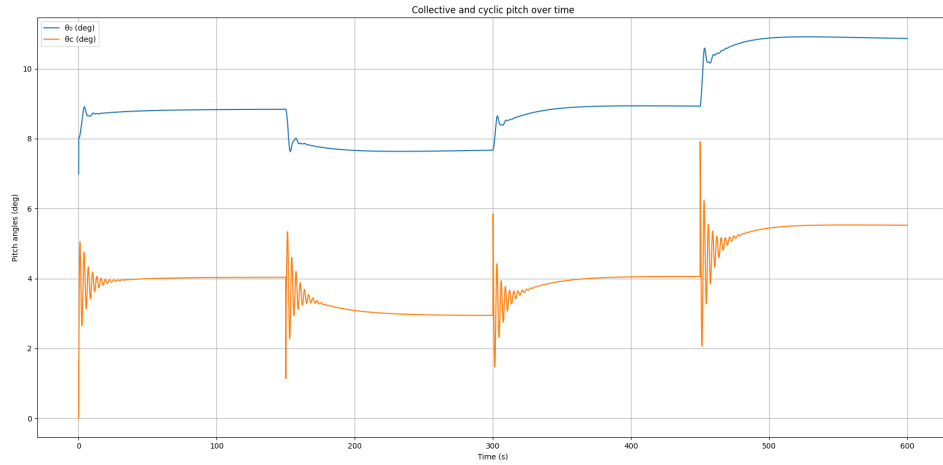


Figure 3.13: Collective and cyclic pitch inputs. The cyclic input compensates for pitch control, while the collective pitch tracks vertical dynamics.

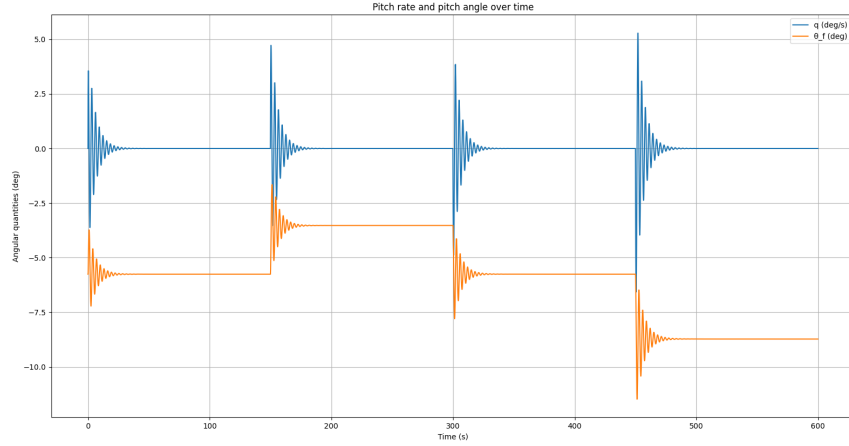


Figure 3.14: Pitch rate q and fuselage pitch angle θ_f during the manoeuvre. Each transition to a new trim condition results in a characteristic pitch motion.

Forward Velocity Tracking. Figure 3.15 shows the actual forward velocity V compared to the desired reference V_{ref} . While the helicopter and controller are able to generally follow the manoeuvre, there is noticeable overshoot. These discrepancies suggest that while the controller is functional, the system exhibits limitations in speed tracking performance. A more detailed analysis of this will follow in the discussion.

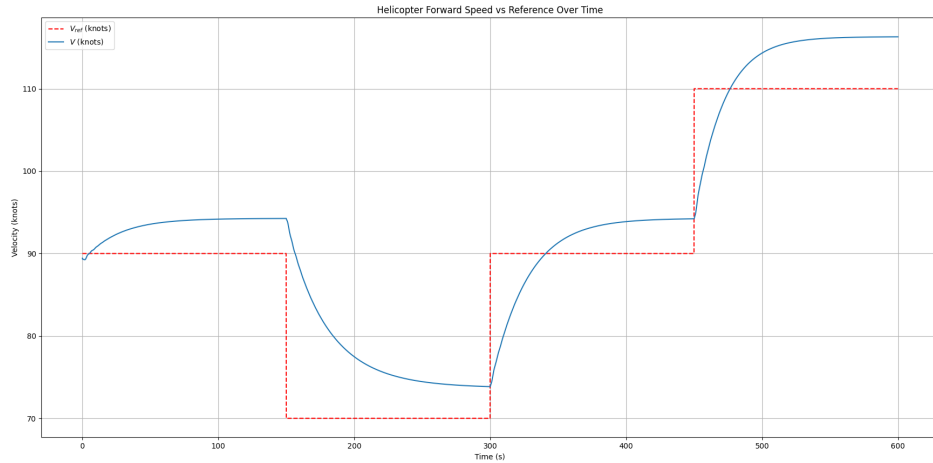


Figure 3.15: Forward velocity tracking. The general trend matches the reference profile, but overshoot and settling delays are present.

In summary, the controller meets the ADS-33 altitude objective with high precision. However, velocity tracking shows room for improvement. In the next section, we will analyze the causes of these deviations and explore possible refinements to the controller.

3.4 Discussion

In this section, we reflect on the performance of the implemented controller and highlight a key discrepancy between the trim conditions computed in Q3.1 and the helicopter model used in Q3.2. While the overall control strategy demonstrates solid altitude tracking and acceptable velocity tracking, we observe a consistent mismatch between the expected and actual system behavior when applying the trim states.

For instance, using the relations found in section 3.2 and $q = 0$, the trim states are defined as:

$$x_{\text{trim}} = \begin{bmatrix} V_{\text{trim}} \cos(\theta_{f,\text{trim}}) & V_{\text{trim}} \sin(\theta_{f,\text{trim}}) & 0 & \theta_{f,\text{trim}} & v_{i,\text{trim}} \end{bmatrix}^T$$

This implies that, if correctly defined, applying these trim values as initial conditions and using the corresponding control inputs should result in no change in the state, such that all derivatives $\dot{x}$ are zero. However, as shown in Figure 3.16, this is not the case.

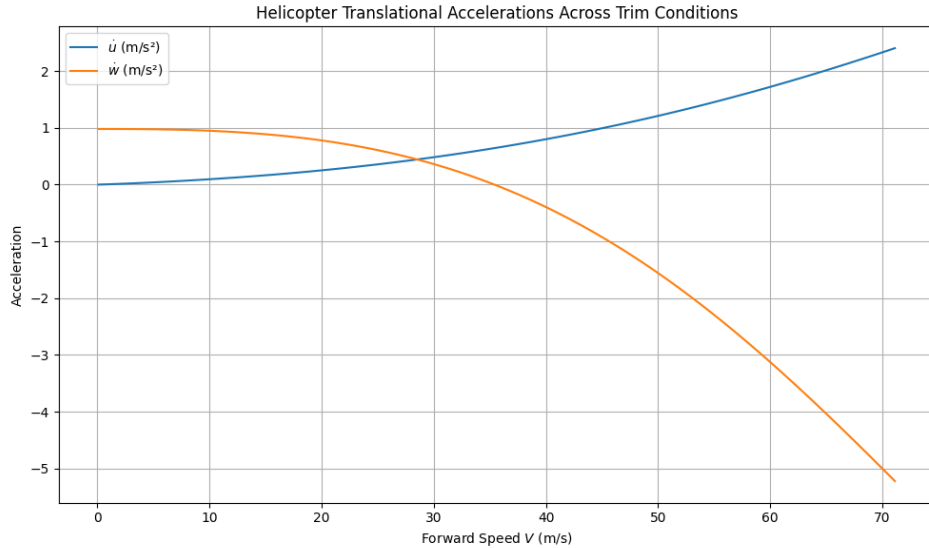


Figure 3.16: Range of trim speeds, where the initial state has been defined by the trim condition. $\dot{u}$ and $\dot{w}$ are the corresponding first change in state. These should be zero.

The nonzero values for the surge and heave accelerations indicate that the helicopter model does not exactly yield equilibrium when initialised at the computed trim state. This reveals a subtle bias in the model or trim assumptions. Since the controller relies on accurate fuselage pitch angles (θ_f) associated with trim velocities, this discrepancy introduces a tracking bias: the reference $\theta_{f,\text{ref}}$ used by the controller does not correspond exactly to the velocity it is trying to achieve. As a result, the helicopter tends to stabilise at a slightly higher speed than intended.

To verify that the trim computation itself is not at fault, Q3.1 was re-solved using an alternative approach, as can be seen in `Q3.1.Validation.py`. The equations for a_1 and C_T were reformulated in matrix form:

$$\begin{pmatrix} 1 + \frac{3}{2}\mu^2 & -\frac{8}{3}\mu \\ -\mu & \frac{2}{3} + \mu^2 \end{pmatrix} \begin{pmatrix} a_1 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} -2\mu^2 \frac{D}{W} - 2\mu\lambda_i \\ \frac{4C_T}{\sigma c_{l\alpha}} + \mu \frac{D}{W} + \lambda_i \end{pmatrix} \quad (3.38)$$

This solution, obtained using `np.linalg.solve`, reproduced the same trim conditions as the original implementation. Therefore, it can be concluded with reasonable certainty that the error is not caused by the trim solver. Despite various attempts to isolate the issue, the root cause of the discrepancy remains unresolved and is left for future work.

Nevertheless, we can quantify and compensate for the bias empirically. As shown in Figure 3.15, the helicopter tends to stabilize at slightly higher speeds than desired. Since the mapping between forward speed and θ_f is one to one, this speed bias can be expressed in terms of a pitch angle correction. The following table shows the approximate bias for each reference speed, determined through trial and error:

Table 3.3: Observed fuselage pitch angle bias at trim

Reference Speed (knots)	Bias (degrees)
70	0.4
90	0.5
110	0.9

Using a least-squares estimation approach, a linear correction relation of the form:

$$\Delta\theta_f = aV_{\text{ref}} + b \quad (3.39)$$

can be fitted to these data points. Resulting in $a = 0.0125$ and $b = -0.525$. Substituting this correction into the controller by adjusting the reference pitch angle:

$$\theta_{f,\text{ref}} = \theta_{f,\text{ref}} + \Delta\theta_f \quad (3.40)$$

effectively compensates for the bias. The resulting improvement in velocity tracking as can be seen in Figure 3.17.

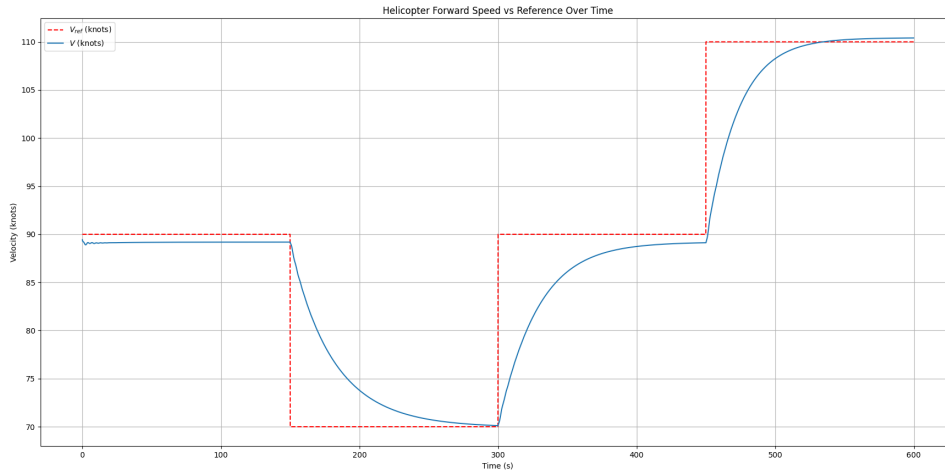


Figure 3.17: Velocity tracking response using bias correction

In conclusion, although the original controller does not perfectly achieve the desired velocities, it successfully maintains stable flight and altitude, and the source of the velocity tracking bias has been identified and partially addressed. Using this correction, the **ADS-33 objective has been achieved**.

Chapter 4

Part 4 - Stability

System matrix determination

In order to analyse the eigenmotion of the helicopter, first, the system matrix A must be set up. This is done by numerical integration using the python module sympy.

First, the trim conditions are determined for the horizontal flight at 70 kts and MTOW. The trim values for the collective and longitudinal pitch are known from section 3.2 and will remain constant throughout the phugoid motion. With these parameters known, all initial values could be determined according to the equations described in section 3.3.

Next, the stability derivatives could be determined by numerical integration. To do this, a system with variables u , w , q , θ_f , λ_i , μ and λ_c . With this system set up, variable equations were set up for the Thrust, Drag and a_1 as a function of these parameters according to Equation 3.25, Equation 1.13 and Equation 3.15 respectively. Now that this dependency was set up, the partial derivatives of X and M could be determined with respect to the state variables. Since the phugoid motion neglects the heave (w) component, this differentiation was not performed. The equations for X and M are given by Equation 4.1 and Equation 4.2

$$X = -mg \sin(\theta_f) - D_{fus} \cos(\theta_f) + T \sin(\theta_c - a_1) \quad (4.1)$$

$$M = -Th \sin(\theta_c - a_1) \quad (4.2)$$

These equations can now be solved for the initial trim conditions, thus returning the stability derivatives for the initial condition. It is assumed that during the phugoid motion, the system matrix does not change, and thus the stability derivatives in trim are assumed to remain constant throughout the motion. The values of the relevant stability derivatives are as follows:

$$X_u = -116.043 \quad ; \quad M_u = 31.30352 \quad ; \quad M_q = -1152.681$$

With these derivatives known, the system matrix can be set up from the simplified

equations of motion for the phugoid:

$$\begin{cases} \dot{u} = (X_u/m)u - g\theta_f \\ \dot{q} = (M_u/I_y)u + (M_q/I_y)q \\ \dot{\theta}_f = q \end{cases}$$

Upon linearization of the state variables, the following model ensues:

$$\begin{bmatrix} \Delta \dot{u} \\ \Delta \dot{q} \\ \Delta \dot{\theta}_f \end{bmatrix} = \begin{bmatrix} \frac{X_u}{m} & 0 & -g \\ \frac{M_u}{I_y} & \frac{M_q}{I_y} & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta q \\ \Delta \theta_f \end{bmatrix} = A \cdot \bar{x} \quad (4.3)$$

Eigenvalue analysis

The system matrix A can thus be calculated with the calculated stability derivatives, the helicopter mass and the moment of inertia as calculated in section 3.1. Filling in all values results in the following A matrix:

$$A = \begin{bmatrix} -0.03988 & 0 & -9.80665 \\ 0.00545 & -0.2008 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

From this matrix, the eigenvalues λ could be determined by solving $\det(A - \lambda I) = 0$. The following eigenvalues were found and are plotted graphically in Figure 4.1

$$\lambda_1 = -0.4679 \quad ; \quad \lambda_2 = 0.1136 + 0.3184i \quad ; \quad \lambda_3 = 0.1136 - 0.3184i$$

From the eigenvalues, it becomes apparent that the phugoid motion consists of a stable,

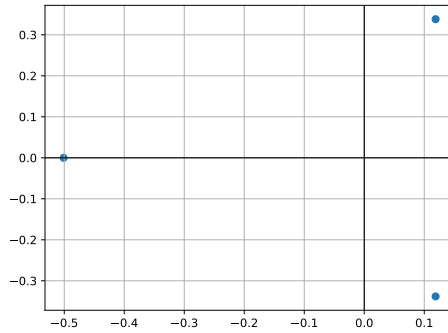


Figure 4.1: Eigenvalues of the phugoid motion at MTOW and 70 kts.

non-oscillatory mode and a conjugate pair of unstable oscillatory modes. The period of the phugoid motion is defined by the two oscillatory modes which each have a period of:

$$T = \frac{2\pi}{|\text{Im}(\lambda)|} = \frac{2\pi}{0.33806} = 19.73s$$

The period obtained from the eigenvalues therefore, corresponds to the expected range of the phugoid period, which is usually around 20 seconds. Secondly, the damping factor of the motion can also be calculated using:

$$\lambda = a + bi \quad ; \quad \zeta = \frac{-a}{\sqrt{a^2 + b^2}}$$

With this, the damping factor of the conjugate pair is found to be $\zeta = -0.336$. This value is negative and thus indicates a negative mode, which is also expected from the helicopter phugoid.

The real, negative eigenvalue indicates a strongly damped, stable mode which over time becomes dominated by the complex conjugate pair. This pair thus defines the long-term phugoid motion and therefore the period and the damping factor of the phugoid motion are:

$$T = 19.73s \quad \zeta = -0.336$$

Finally, the model described by Equation 4.3 was simulated from. The resulting values for Δu , Δq and $\Delta \theta_f$ are plotted against time in Figure 4.2. The results from this figure are the result of a perturbation in the forward velocity (Δu) of 0.1 m/s. Here, it can also be seen that the change in forward velocity changes periodically and increases in amplitude, thus indicating static stability and dynamic instability. By analysing the difference in peaks in Δu , the period could also be determined to be 18.63 seconds. This is extremely close to the period calculated from λ_2 and λ_3 and thus shows that the assumption that modes 2 and 3 dominate the phugoid is accurate.

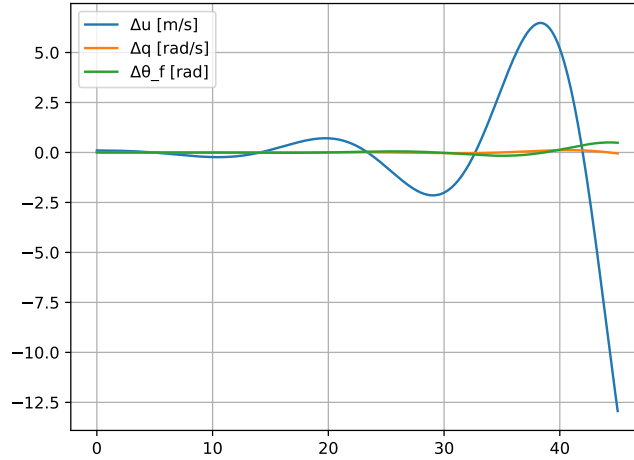


Figure 4.2: The timewise evolution of the velocity, pitch rate and pitch angle after a velocity perturbation of 1 m/s.

Use of ChatGPT

During the making of this assignment, we have made some use of AI tools (mainly ChatGPT). We made sure to check all the information given by it, and no piece of information was taken as the absolute truth. The main way in which it has been used is to write pieces of code such as the making of certain plots or when making use of unfamiliar libraries. Some use was also made to find information regarding the specific parameters of the helicopter, in these cases we only used it to find the relevant sources and then use those to find the required information. Finally, chatGPT in combination with Wolfram Alpha has been used to check some of the calculations, such as integrations and differentiations. These have however never been directly used in the making of this assignment, only as a check.

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